Lecture 3
Internet History, Network Core

Warm-up questions

Q1: Which of the following best describes an access network?

a) The collection of routers that forms the Internet core

b) The network that connects an end system to the first/edge router

c) A network that connects only servers in a data center

d) A network that connects different global ISPs together

Warm-up questions

Q2: Your phone is on 5G, and your computer is connected to the Internet via Ethernet. Both load the same webpage from the same server (e.g., www.google.com).

a) Can you name one thing that is identical about the two connections?

b) How about one thing that is different about the two connections?

Warm-up questions

Q3: Suppose there are *N* access ISPs. Why is connecting every ISP directly to every other ISP not a scalable Internet architecture?

a) Routers cannot connect to more than one ISP.

b) Every ISP must use a different protocol.

c) The number of required interconnections grows approximately as *O(N^2).*

d) Access ISPs are not allowed to communicate directly.

Warm-up questions

Q4: In the Internet's "network of networks," architecture what is an IXP?

a) A protocol for finding end-to-end paths

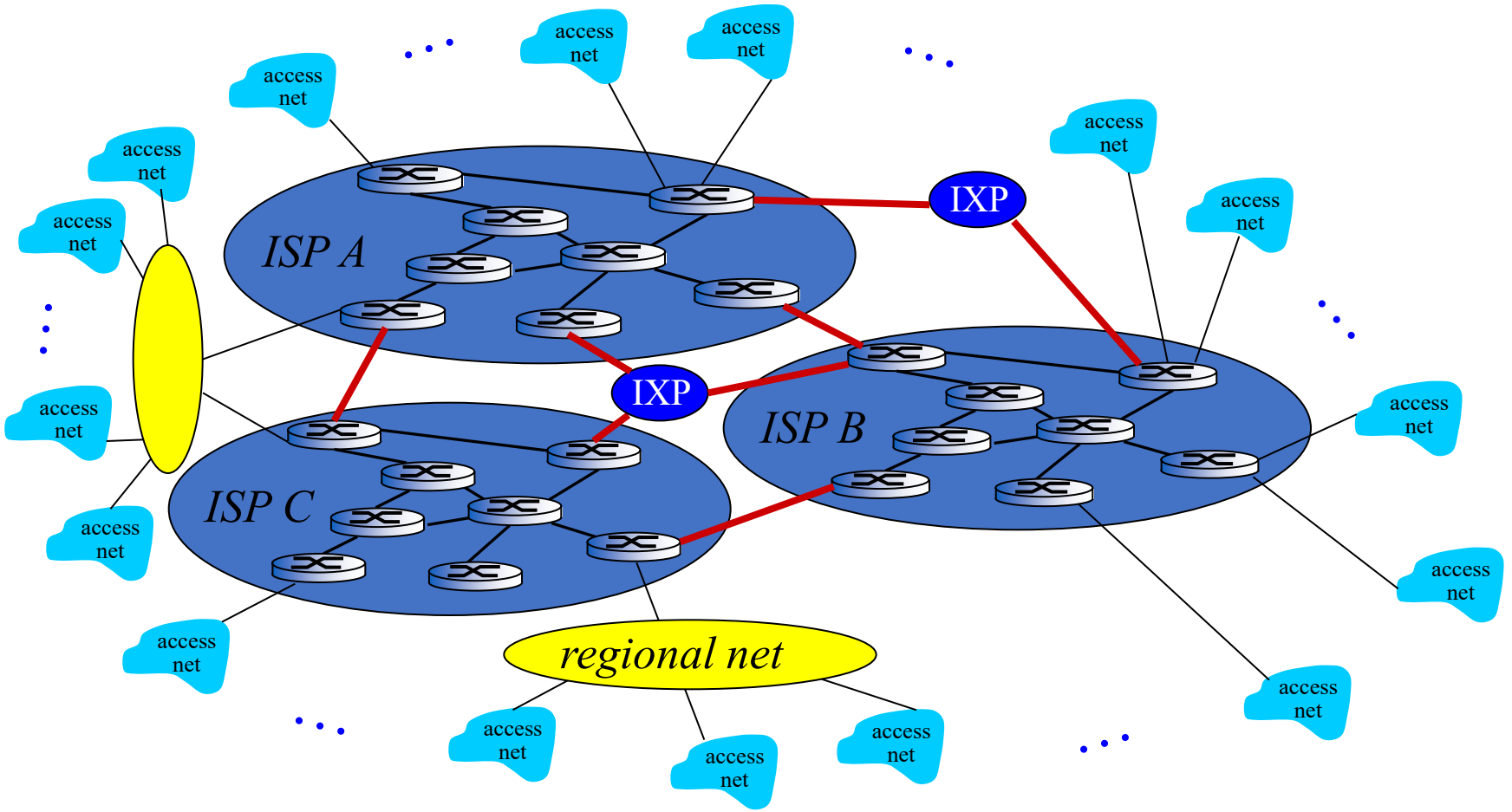
b) A physical location where ISPs interconnect and exchange traffic

c) A type of access network for homes

d) The addressing scheme replacing IPv4

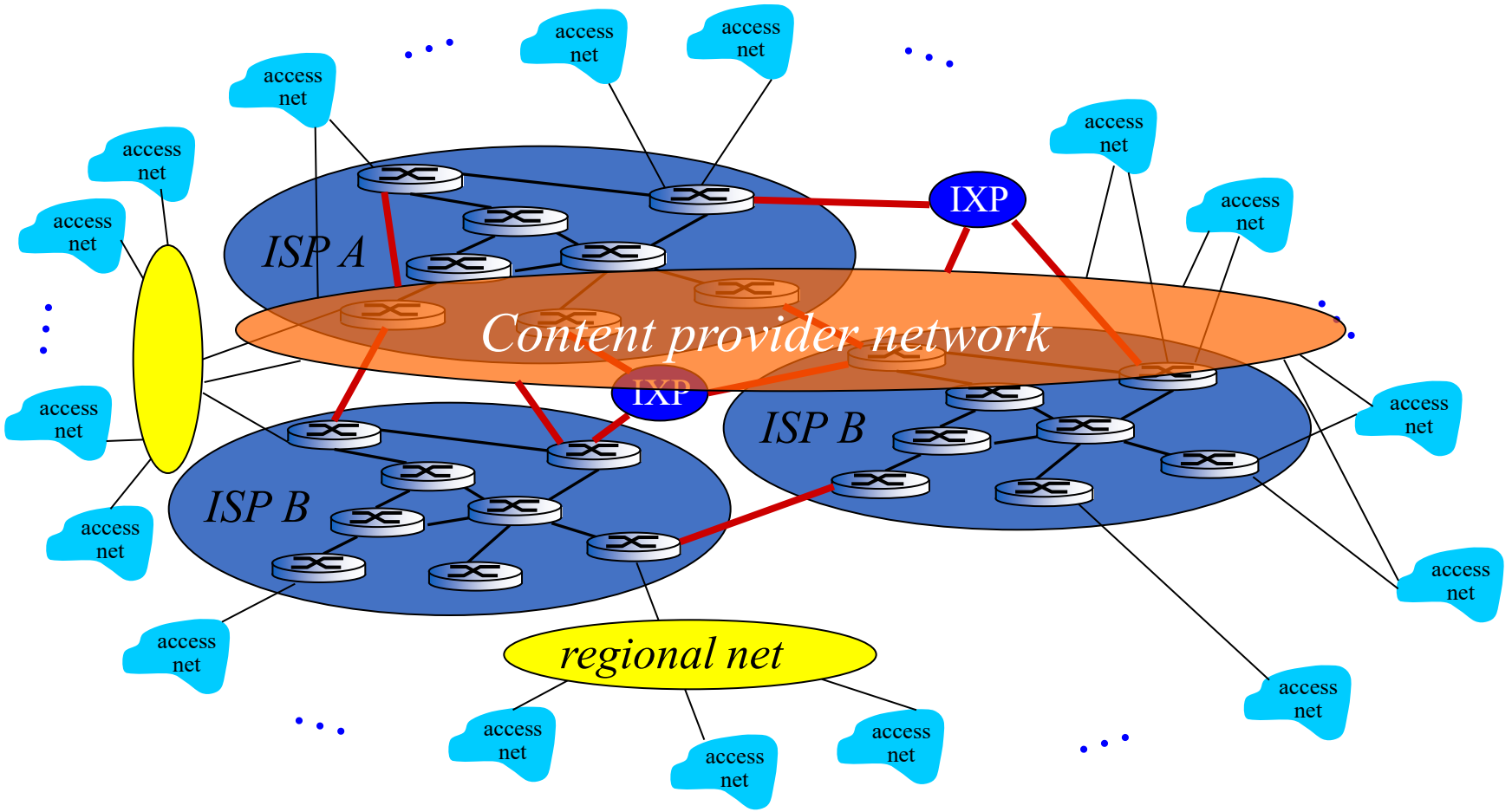
Internet Structure: Network of Networks

… and regional networks may arise to connect access nets to ISPS

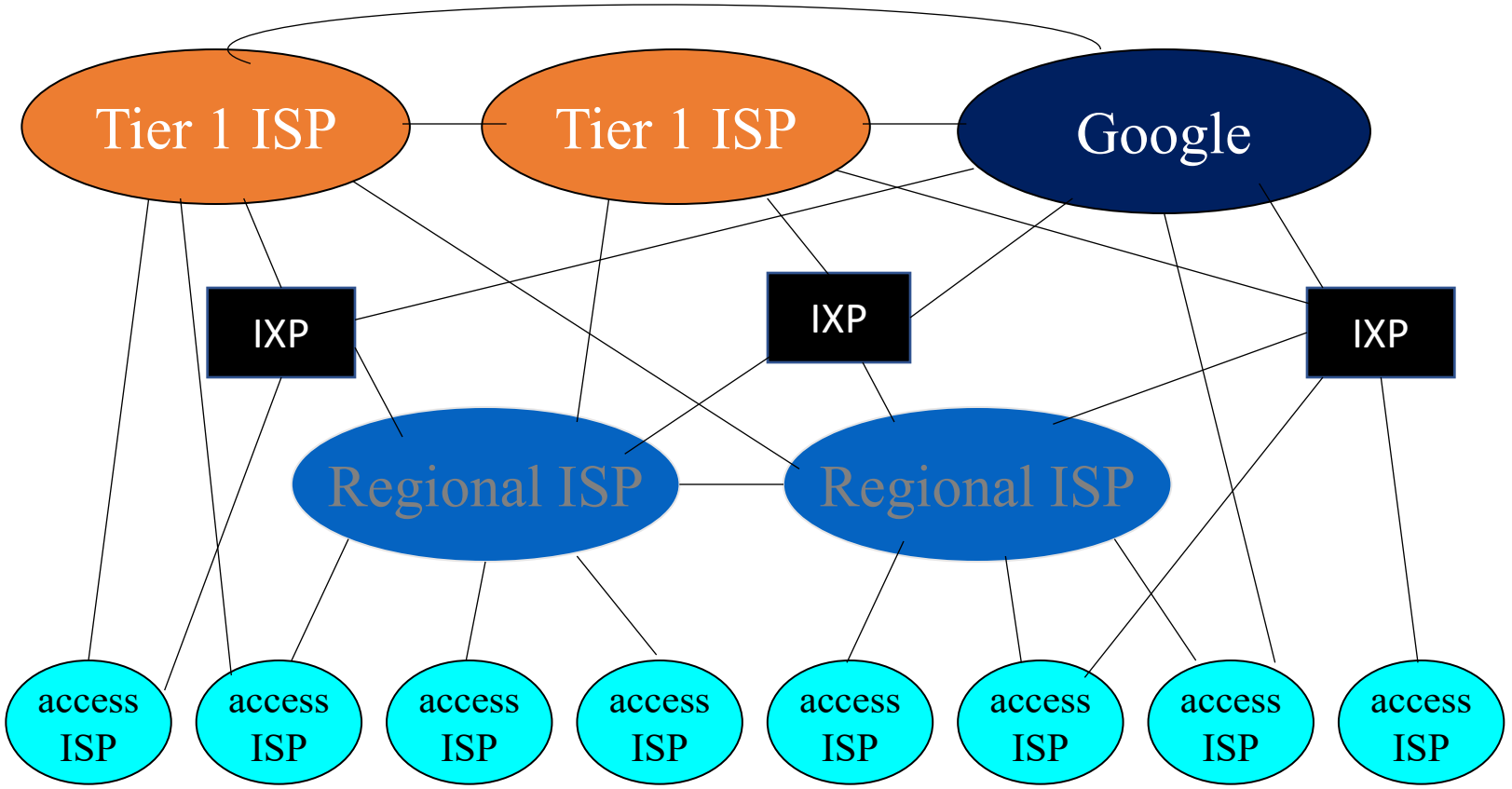


Internet Structure: Network of Networks

… and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Internet Structure: Network of Networks

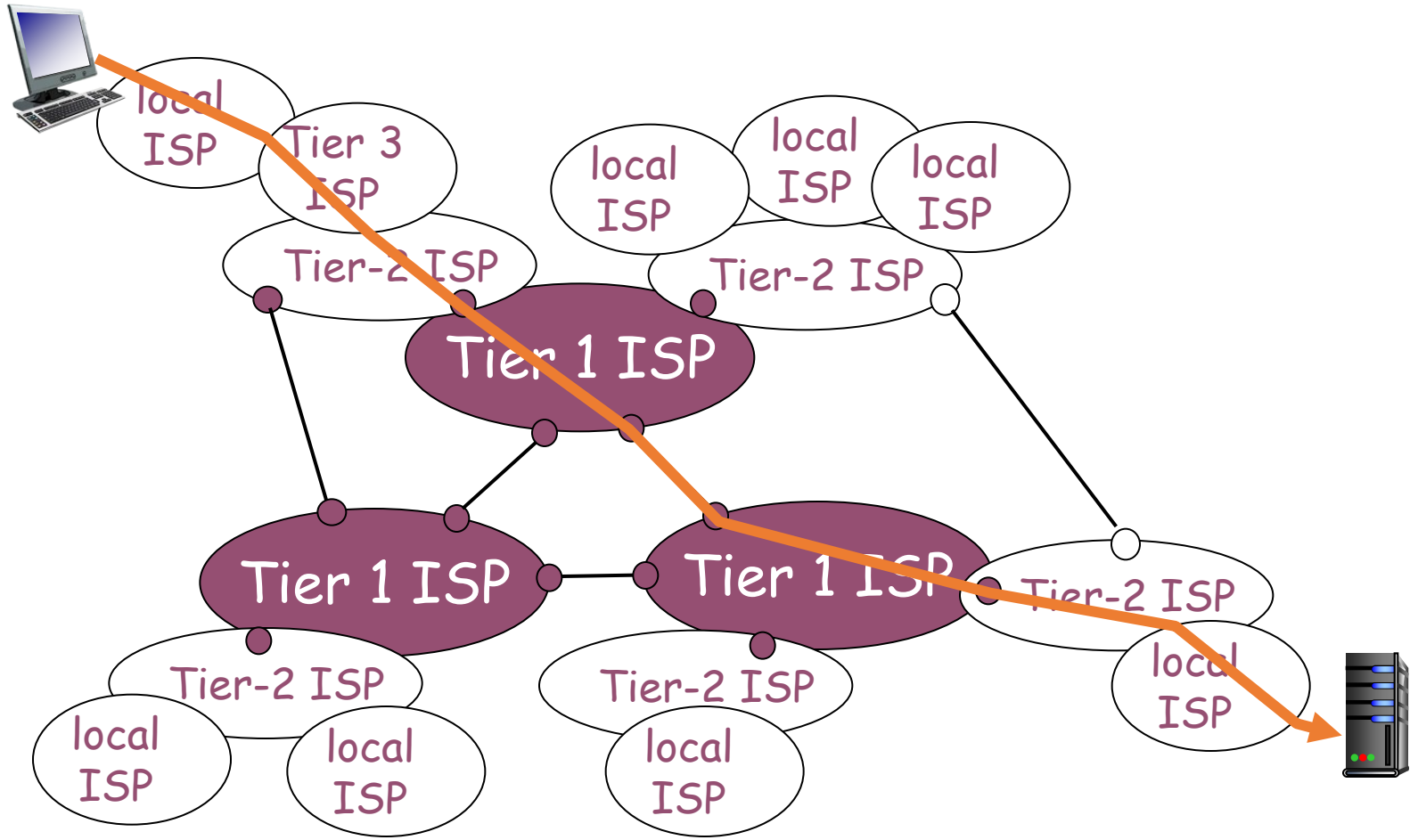


At “center”: small # of well-connected large networks

- “tier-1” commercial ISPs (AT&T, Verizon, Sprint, NTT, Singtel, PCCW, Telstra, Deutsche Telekom and British Telecom), national & international coverage
- content provider networks (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

Internet Structure: Network of Networks

- A packet passes through many networks!

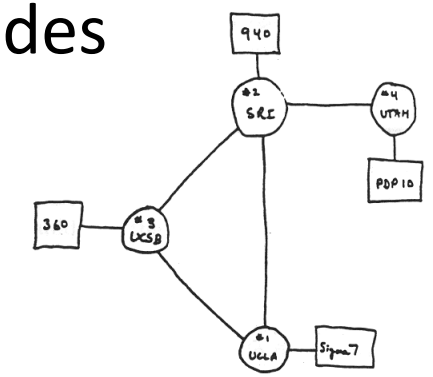


Internet history

1961-1972: Early packet-switching principles

- 1961: Kleinrock - queueing theory shows effectiveness of packet-switching
- 1964: Baran - packet-switching in military nets
- 1967: ARPAnet conceived by Advanced Research Projects Agency
- 1969: first ARPAnet node operational

- 1972:
  - ARPAnet public demo
  - NCP (Network Control Protocol) first host-host protocol
  - first e-mail program
  - ARPAnet has 15 nodes



THE ARPA NETWORK

Internet history

1972-1980: Internetworking, new and proprietary nets

- 1970: ALOHAnet satellite network in Hawaii
- 1974: Cerf and Kahn - architecture for interconnecting networks
- 1976: Ethernet at Xerox PARC
- late70's: proprietary architectures: DECnet, SNA, XNA
- late 70's: switching fixed length packets (ATM precursor)
- 1979: ARPAnet has 200 nodes

Cerf and Kahn's internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best-effort service model
- stateless routing
- decentralized control

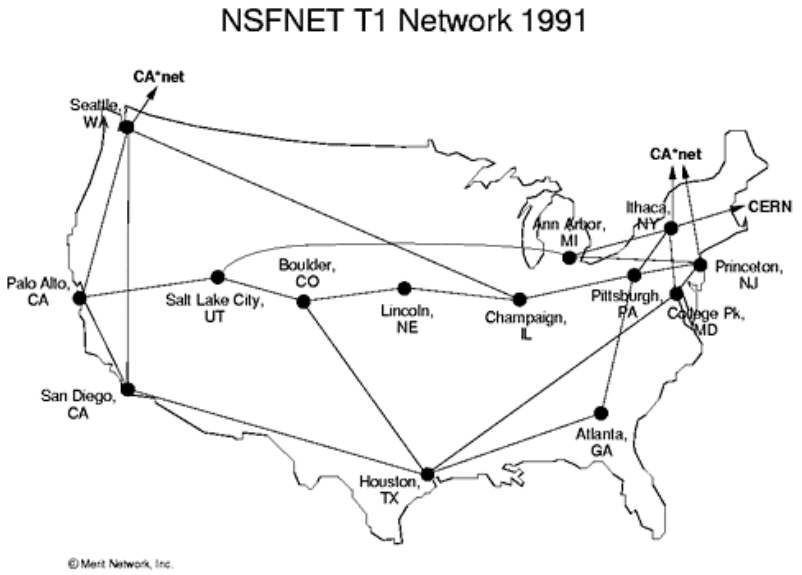
define today's Internet architecture

Internet history

1980-1990: new protocols, a proliferation of networks

- 1983: deployment of TCP/IP
- 1982: SMTP e-mail protocol defined
- 1983: DNS defined for name-to-IP-address translation
- 1985: FTP protocol defined
- 1988: TCP congestion control

- new national networks: CSnet, BITnet, NSFnet, Minitel
- 100,000 hosts connected to confederation of networks



Internet history

1990, 2000s: commercialization, the Web, new applications

- early 1990s: ARPAnet decommissioned
- 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- early 1990s: Web
  - hypertext [Bush 1945, Nelson 1960's]
  - HTML, HTTP: Berners-Lee
  - 1994: Mosaic, later Netscape
  - late 1990s: commercialization of the Web

late 1990s – 2000s:

- more killer apps: instant messaging, P2P file sharing
- network security to forefront
- est. 50 million host, 100 million+ users
- backbone links running at Gbps

Internet history

2005-present: more new applications, Internet is "everywhere"

- rise of smartphones (iPhone: 2007)
- aggressive deployment of broadband access
- increasing ubiquity of high-speed wireless access: 4G/5G, WiFi
- emergence of online social networks:
- service providers (e.g., Google) create their own networks
  - bypass commercial Internet to connect "close" to end user, providing "instantaneous" access to search, video content, …
- enterprises run their services in "cloud" (e.g., Amazon Web Services, Microsoft Azure)

Chapter 1: roadmap

- Basics of Network Building Blocks and Protocols
- Internet Structure
- Network Core: Packet vs. Circuit Switching
- Network Performance: Delay, Packet loss, Throughput
- Protocol Layers and Service Models

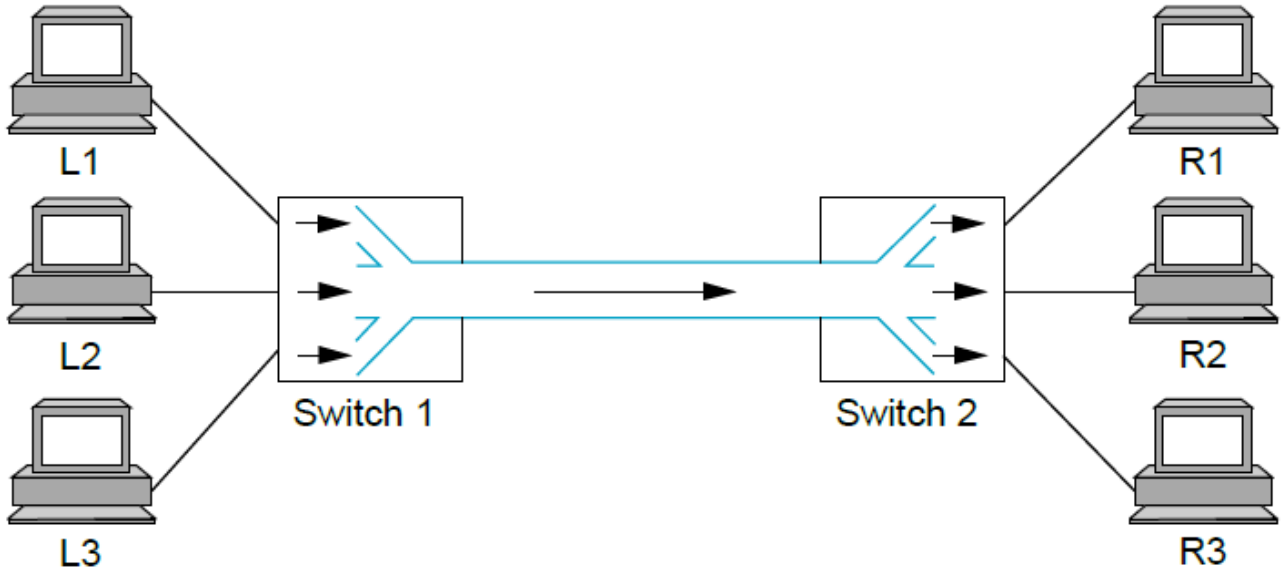
Network Resources



- Network resources are limited ➔ Need to be managed carefully
- Bandwidth (Link Capacity): Data rate per unit of time (bits/sec, b/s, bps)
  - 1 Kb/s = 10^3 b/s (modem, 2G cell phones)
  - 1 Mb/s =10^6 b/s (Ethernet, coax, twisted pair, Wi-Fi, 3G/4G cell phones)
  - 1 Gb/s = 10^9 b/s (Gigabit Ethernet, optical networks)
  - 1 Tb/s = 10^{12} b/s (Submarine communication cables)
- Memory (Buffer Capacity): Expressed in bytes (1 byte = 8 bits)
  - 1 KB = 10^3 bytes (RF-IDs, sensors)
  - 1 MB =10^6 bytes (Cache memory)
  - 1 GB = 10^9 bytes (RAM, Flash memory)
  - 1 TB = 10^{12} bytes (Hard Disk, Magnetic disk)

Operation at the Network Core

- Multiplexing: Sharing network resources among multiple users



Operation at the Network Core

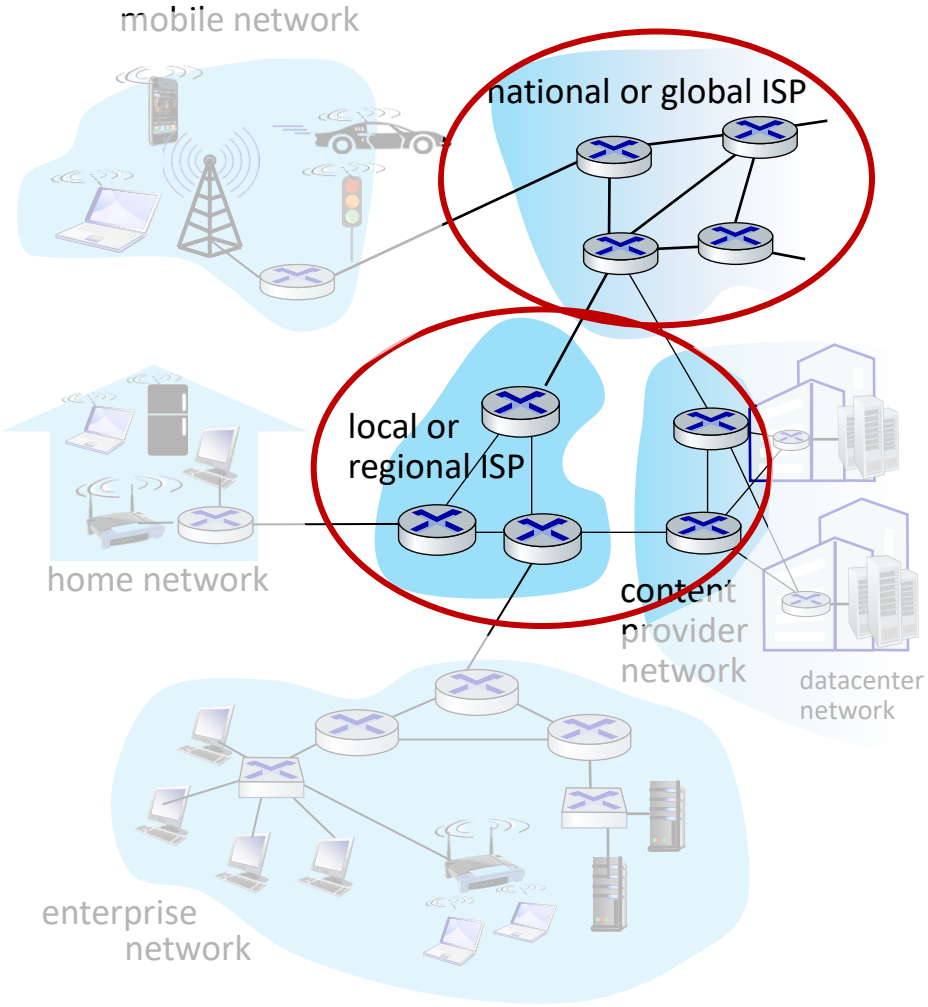
- Multiplexing: Sharing a resource among multiple users
  - Frequency Division Multiplexing (FDM): partition resource in frequency
  - Time Division Multiplexing (TDM): partition resource in time
  - Statistical multiplexing: Allow access on demand

Two Types of Network Core Architectures

Fundamental question:

How is data transferred through network?

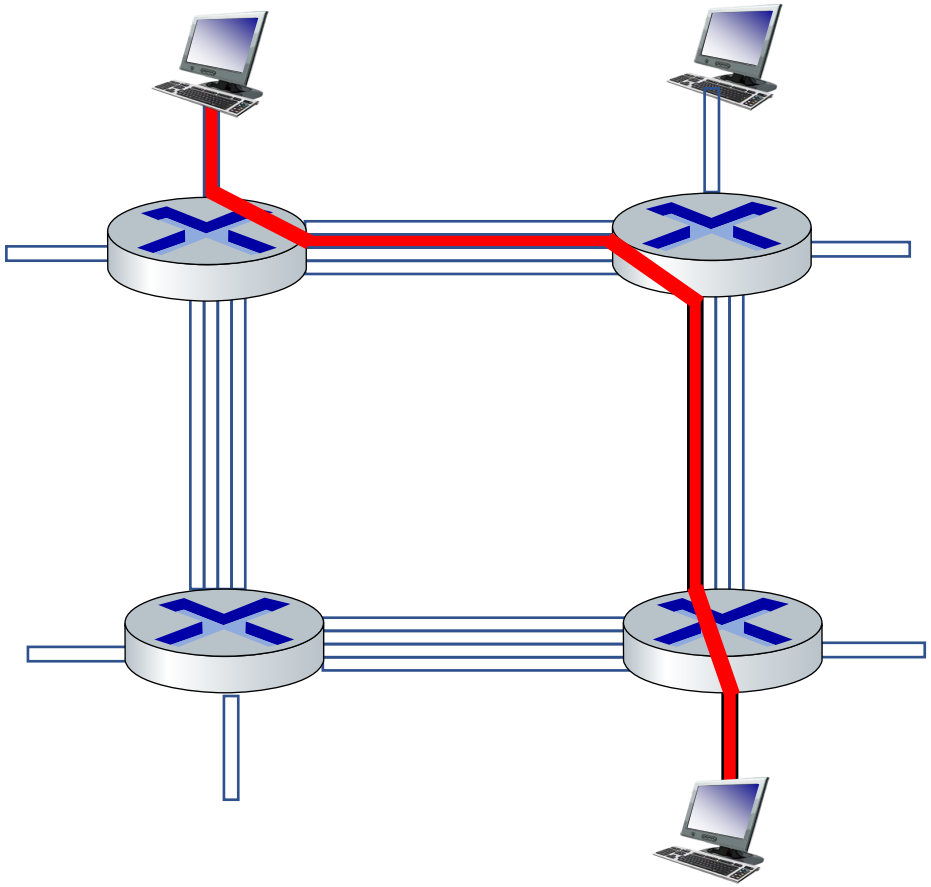
- Circuit switching: dedicated resource allocation
- Packet switching: data sent through network in discrete chunks



Network Core: circuit switching

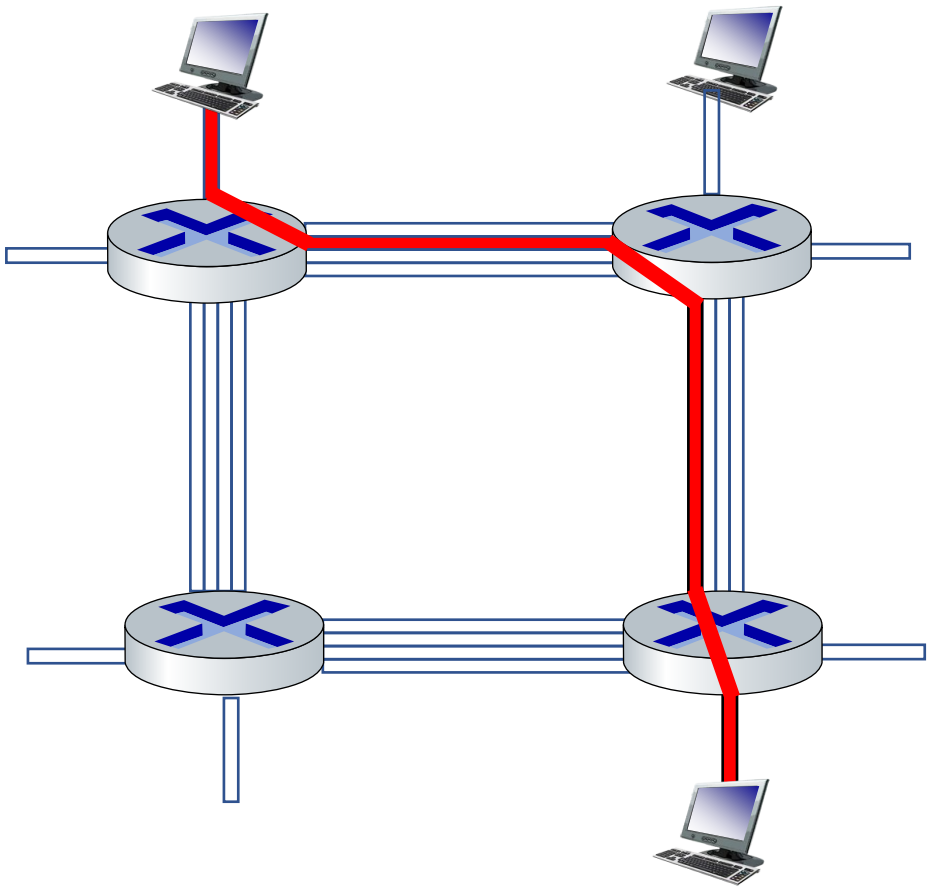
End-end resources allocated to, reserved for "call" between source and destination

- dedicated resources: no sharing
- Rigid allocation of resources (no sharing)
- Inefficient for bursty data traffic (web traffic, etc.)
  - ***Underutilized*** due to periods of inactivity



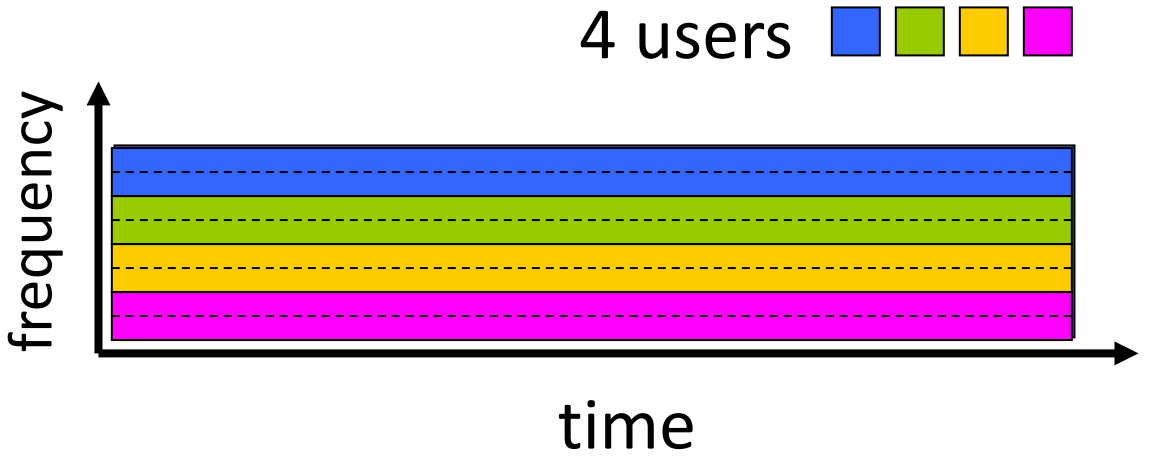
Network Core: circuit switching

- Network resources (e.g., bandwidth) divided into "pieces"
- Pieces allocated to calls
- Dividing link bandwidth into "pieces"
  - Frequency division
  - Time division

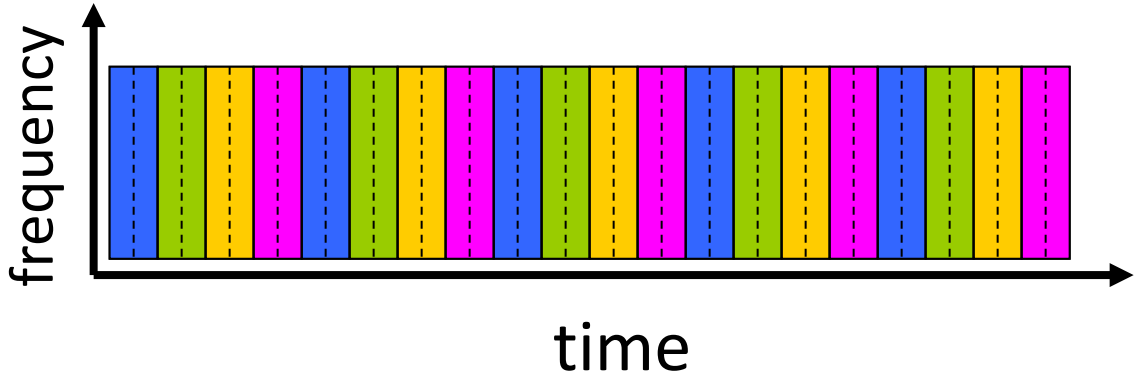


Circuit switching: FDM and TDM

Frequency Division Multiplexing (FDM)

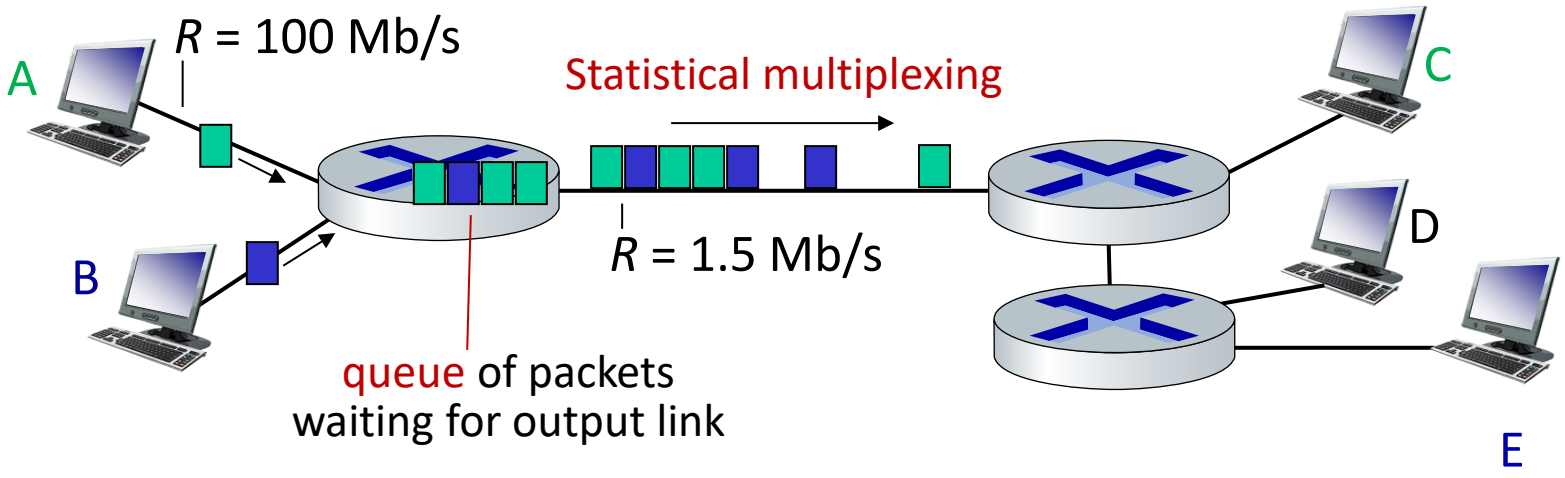


Time Division Multiplexing (TDM)

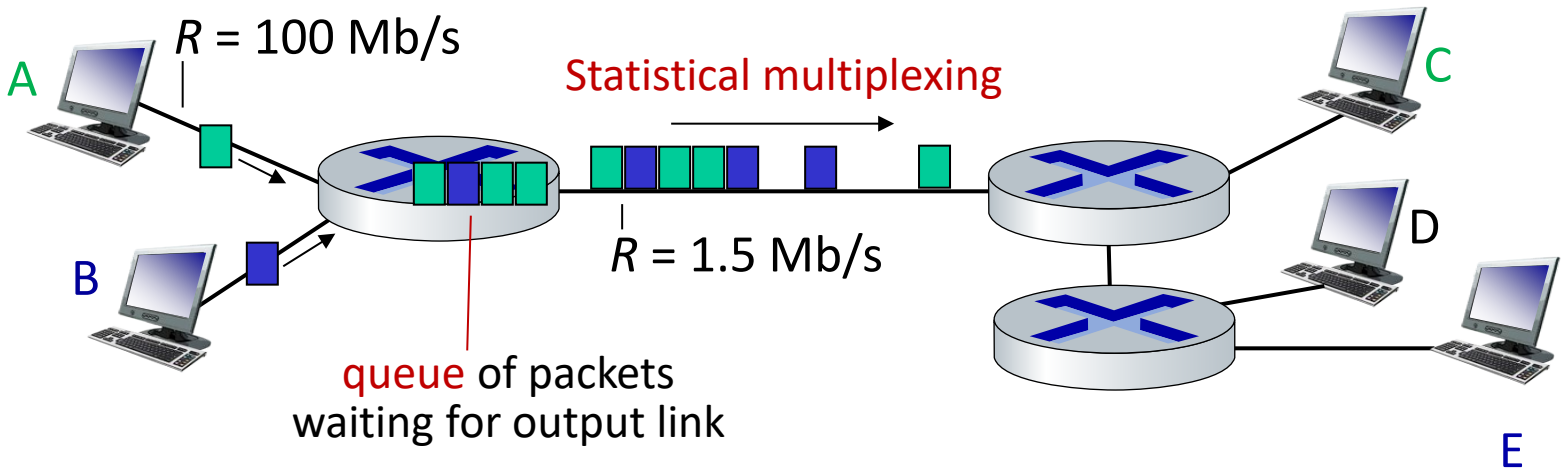


The network core: Packet Switching

- Packet-switching: hosts break application-layer messages into *packets*
  - Users A and B packets *share* network resources
  - Each packet uses full link bandwidth
  - Resources used as needed



Packet Switching: Statistical Multiplexing



- Danger!
  - Aggregate resource demand can exceed amount available
  - Congestion: Packets queue; Wait for link availability

Examples

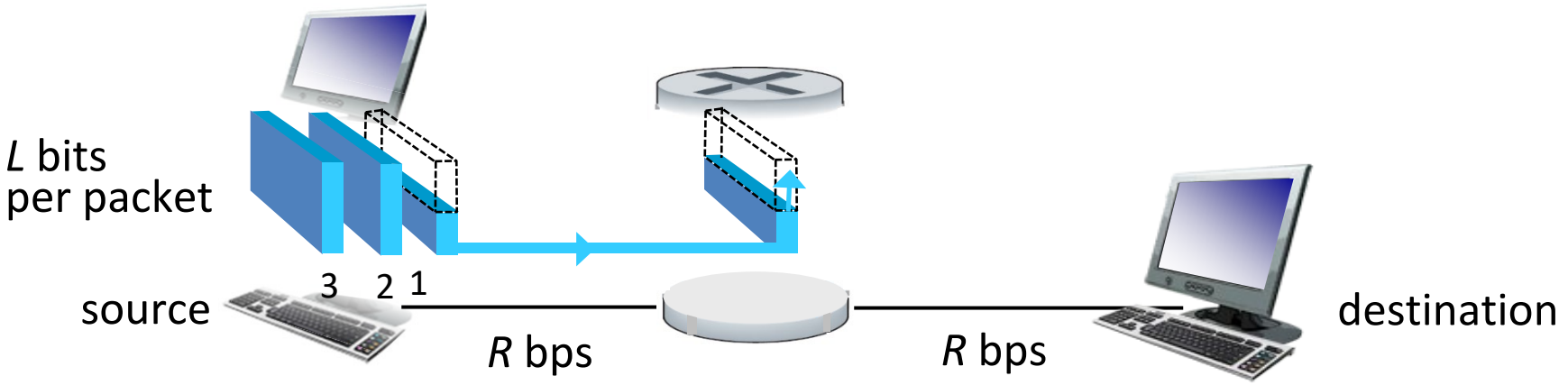
- Circuit Switching Networks
  - Traditional Landline Telephone Calls ➔ a dedicated circuit between your phone and the receiver's phone.
  - Early Cellular Networks (1G, some 2G)
- Packet Switching Networks
  - Modern VoIP calls (Skype, WhatsApp, etc.)
  - Internet network
  - Cellular network (e.g., VoLTE)

Packet switching: Pros and Cons

- Great for “bursty” data – sometimes has data to send, but at other times not
  - resource sharing
  - simpler, no call setup

- Resource contention may lead to congestion:
  - aggregate resource demand can exceed capacity
  - packets queue (are buffered), wait for link use
  - excessive congestion: packet delay and loss
  - protocols needed for reliable data transfer, congestion control, fairness for network users

Packet-switching: store-and-forward



- *Store and forward:* *entire* packet must arrive at router before it can be transmitted on next link

Packet Switching Features

Question: Which of the characteristics below are associated with ***packet switching***?

A. Data may be queued before being transmitted due to other user's data
B. Frequency Division Multiplexing (FDM) and Time Division Multiplexing (TDM) are two approaches for implementing this technique.
C. Reserves resources needed for a call from source to destination.
D. It is efficient for bursty data traffics